Task 4: Setup and Use a Firewall on Windows/Linux

Objective

Configure and test basic firewall rules to allow or block network traffic using **Windows Defender Firewall** and **UFW (Uncomplicated Firewall)** on Linux.

Tools Used

- Windows Defender Firewall
- UFW (Linux / Kali / Ubuntu)
- Command Prompt / Linux Terminal
- Screenshot Tool

Windows Firewall

1. Open Firewall Settings

Press `Windows + R`, type:

```
wf.msc
```

This opens **Windows Defender Firewall with Advanced Security**.

2. View Existing Rules

- Open **Inbound Rules**
- Review currently active rules

3. Create a Rule to Block Port 23

- Click **New Rule**
- Select **Port**
- Choose **TCP**
- Enter `23`
- Select **Block the connection**
- Apply to all profiles

- Save as **Block Telnet Port 23**

4. Test the Rule

Use Command Prompt:

```
telnet localhost 23
```

Expected result: connection blocked.

5. Remove Test Rule

Delete the created inbound rule to restore the original configuration.

Linux Firewall (UFW)

1. Check Firewall Status

```
sudo ufw status
```

2. Allow SSH Port

```
sudo ufw allow 22
```

3. Enable Firewall

```
sudo ufw enable
```

4. Block Telnet Port

```
sudo ufw deny 23
```

5. Verify Rules

```
sudo ufw status numbered
```

Example Output:

```
[1] 22 ALLOW Anywhere
[2] 23 DENY Anywhere
```

6. Test the Rule

```
telnet localhost 23
```

Expected result: connection refused / blocked.

7. Remove Test Rule

```
sudo ufw delete deny 23
```

8. Disable Firewall

```
sudo ufw disable
```

Summary

This task demonstrates how firewalls filter traffic based on rules.

- **Windows:** blocked inbound Telnet traffic on port 23
- **Linux:** allowed SSH on port 22 and blocked Telnet on port 23
- Tested the rules successfully
- Restored the original configuration after testing

Outcome

Practical understanding of:

- firewall rule creation
- inbound traffic blocking
- port-based filtering
- Windows and Linux firewall administration
- basic network security controls